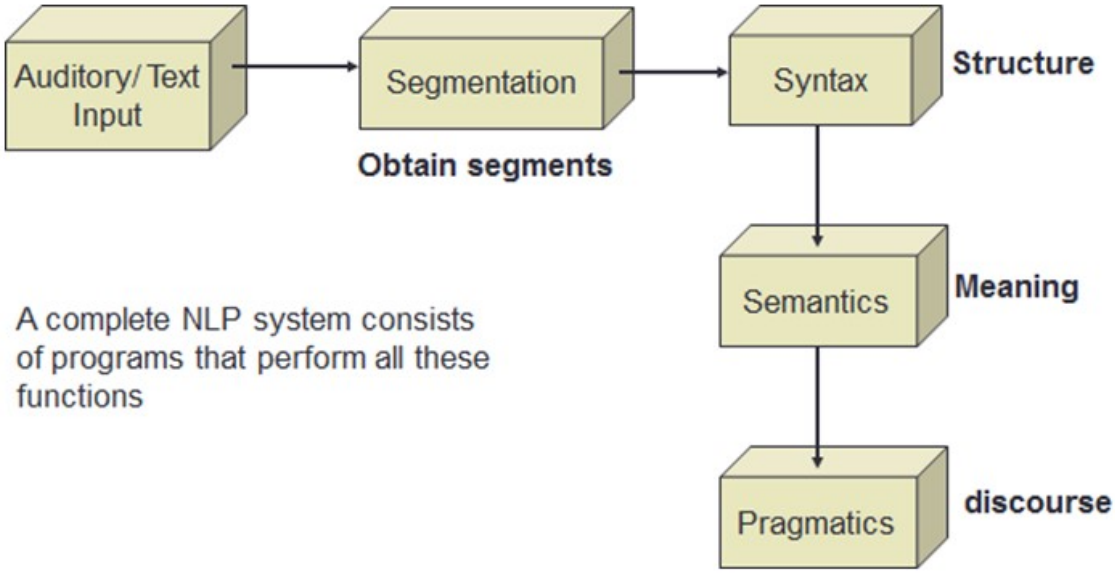
Natural Language Processing

- Language is a means of communication for humans. By studying language, we come to understand more about the world.
- If we can succeed at building a computational model of language, we will have a powerful tool for communicating about the world.
- We look at how we can exploit knowledge about the world, in combination with linguistic facts, to build computational natural language systems.
- Natural Language Processing (NLP) is the process of computer analysis of input provided in a human language (natural language) and conversion of this input into a useful form of representation.
- NLP is one of the fields of AI that processes or analyzes written or spoken language.
- NLP involves processing of speech, grammar, and meaning.
- NLP is composed of two parts: NLU (Natural Language Understanding) and NLG (Natural Language Generation).
- Understanding any language requires detailed knowledge of that language.

NLP Steps/Processes



Input/Source

- The input of an NLP system can be written text or speech.
- Quality of input decides the possible errors in language processing; high-quality input leads to correct language understanding.

Segmentation

- The text inputs are divided into segments (chunks) and the segments are analyzed. Each such chunk is called a frame.

Syntactic Analysis

- Syntactic analysis takes an input sentence and produces a representation of its grammatical structure.
- A grammar describes the valid parts of speech of a language and how to combine them into phrases.
- The grammar of English is nearly context-free.

Grammar: A computer grammar specifies which sentences are in a language and their parse trees. A parse tree is a hierarchical structure that shows how the grammar applies to the input. Each level of the tree corresponds to the application of one grammar rule.

Syntax:

1. sentence ↔ noun_phrase verb_phrase
2. noun_phrase ↔ noun
3. noun_phrase ↔ article noun
4. verb_phrase ↔ verb
5. verb_phrase ↔ verb noun_phrase
6. article ↔ a
7. article ↔ the
8. noun ↔ man
9. noun ↔ dog
10. verb ↔ likes
11. verb ↔ bites

Parse Tree

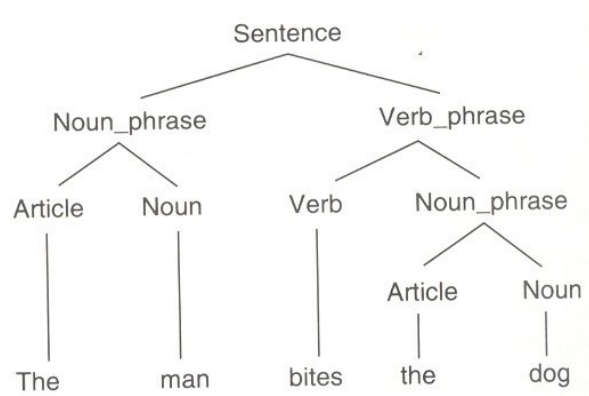


Figure 10.3 Parse tree for the sentence "The man bites the dog."

Semantic Analysis

- Semantic analysis is a process of converting the syntactic representations into a meaning representation.
- This involves the following tasks:
    - **Word Sense Determination**
    - **Sentence-Level Analysis**

Word Sense: Words have different meanings in different contexts.
Example: Mary had a bat in her office.

- bat = "a baseball thing"
- bat = "a flying mammal"

Sentence-Level Meaning: Once the words are understood, the sentence must be assigned some meaning.
Examples:

- She saw her duck.
- I saw a man with a telescope.

Non-examples: "Colorless green ideas sleep furiously." This would be rejected semantically as "colorless green" would make no sense.

Pragmatic Analysis

- Pragmatics comprises aspects of meaning that depend upon the context or facts about the real world.
- These aspects include:
    - Pronouns and referring expressions
    - Logical inferences that can be drawn from the meanings of a set of propositions
    - Discourse structure: the meaning of a collection of sentences taken together.

Examples:

- Jack fell. Jill brought him a band-aid. (Jack got hurt and Jill wanted to help.)
- We got seven letters today. (We got only seven letters and not eight.)
- Understanding toothpaste directions: "For best results, squeeze tube from the bottom and flatten as you go up."

NLP Aids

To understand or analyze language, some aids or tools can be used in NLP.

1. **Lexicon or Dictionary**
    - A lexicon defines the words of a language that a system knows about.
    - This includes common words and words that are specific to the domain of the application.
    - Entries include meanings for each word and its syntactic and morphological behavior.
2. **Morphological Analysis System**
    - Morphological analysis is the process of recognizing the suffixes and prefixes that have been attached to a word.
    - We do this by having a table of affixes and trying to match the input as: prefixes + root + suffixes.
3. **Syntactic Analysis System**
    - Takes an input sentence and produces a representation of its grammatical structure.
    - Grammar (mostly context-free grammar) and parser are used.
4. **Semantic Analysis System**
    - Converts the syntactic representation into a meaning representation.
    - It involves:
        - Word sense determination
        - Sentence-level analysis
        - Knowledge representation
5. **Pragmatic Analysis System**
    - Studies how knowledge about the world and language conventions interact with the literal meaning.
    - Depends upon context and facts about the real world.

Natural Language Analysis Techniques

Two commonly used techniques in natural language processing are:

1. **Keyword Analysis or Pattern Matching**
    - Accept the given sentence as input.
    - Segment the sentence.
    - Identify keywords in each segment.
    - If a keyword is present:
        - If only one keyword is present, give a suitable reply based on the keyword.
        - If more than one keyword is present, prioritize them and give a suitable reply.
    - If no keyword is present, give a random reply.
2. **Syntactic Driven Parsing Technique**

- Words can fit together into higher-level units such as phrases, clauses, and sentences.
- Interpretations of larger groups of words are built up out of the interpretation of their syntactic constituent words or phrases.

NLP Problems

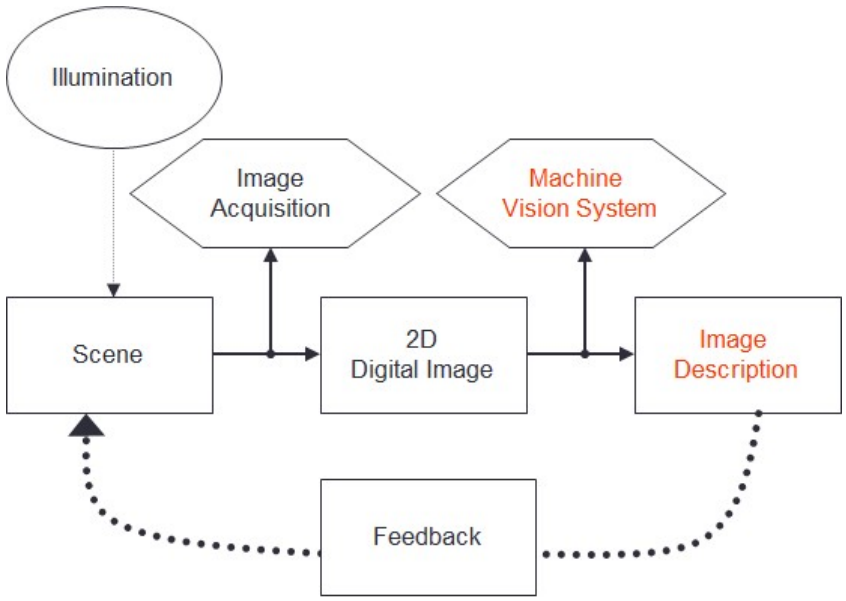
1. The same expression means different things in different contexts.
    - Where's the water? (Chemistry lab? Must be pure.)
    - Where's the water? (Thirsty? Must be drinking water.)
    - Where's the water? (Leaky roof? It can be dirty.)
2. No natural language program can be complete because new words, expressions, and meanings can be generated quite freely.
    - I'll fax it to you.
3. There are lots of ways to say the same thing.
    - Ram was born on October 11.
    - Ram's birthday is October 11.
4. Sentences and phrases might have hidden meanings.
    - "Out of sight, out of mind." → "Invisible idiot."
    - "The spirit was willing but the flesh was weak." → "The vodka was good, but the meat was bad."
5. Problems due to syntax and semantics.
6. Problems due to extensive use of pronouns. (semantic issue)
    - Example: Ravi went to the supermarket. He found his favorite brand of coffee in the rack. He paid for it and left.
7. Use of grammatically incorrect sentences.
    - Example: Her ice eats. (syntax issue)
8. Use of conjunctions to avoid repetition of phrases causes problems in NLP.
    - Example: Ram and Hari went to the restaurant. While Ram had a cup of coffee, Hari had tea.
    - Hari had a cup of tea.

Introduction to Machine Vision

The goal of Machine Vision is to create a model of the real world from images.

- A machine vision system recovers useful information about a scene from its two-dimensional projections.
- The world is three-dimensional.
- Two-dimensional digitized image.

Knowledge about the objects (regions) in a scene and projection geometry is required. The information which is recovered differs depending on the application such as satellite, medical images, etc.



The goal of a machine vision system is to compute a meaningful description of the scene (e.g., object)

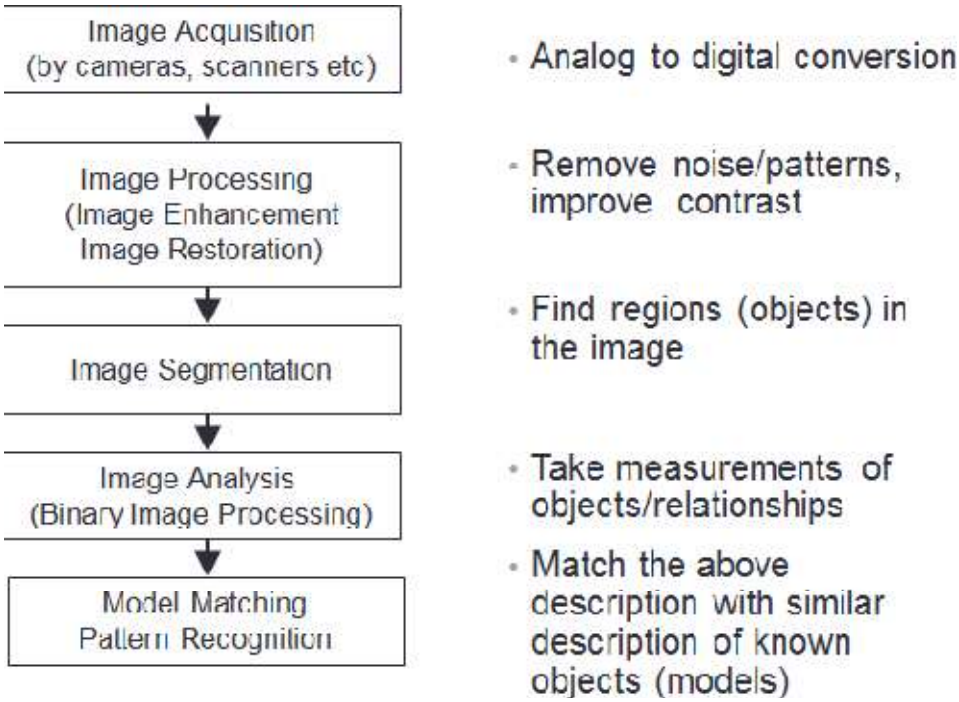
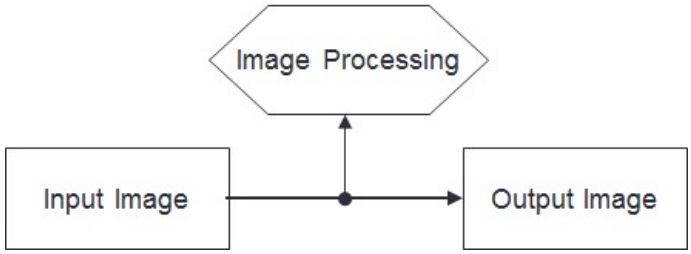


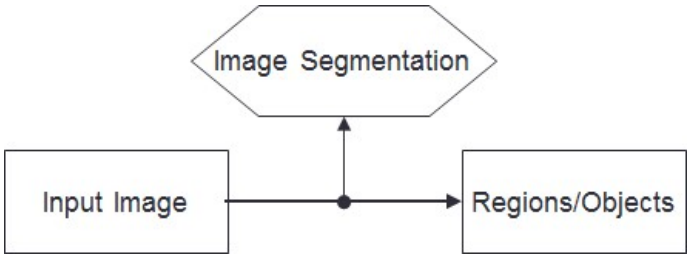
Image Processing



Various procedures such as:

- Image enhancement (filtering, edge detection, surface detection, computation of depth).
- Image restoration (removing point/pattern degradation: there exists a mathematical expression of the type of degradation like added multiplicative noise, sin/cos pattern degradation, etc.).

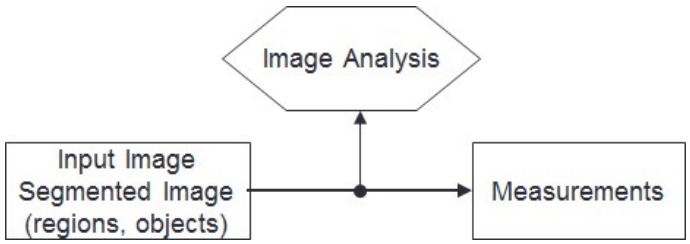
Image Segmentation



Classify pixels into groups (regions/objects of interest) sharing common characteristics:

- Intensity/Color, texture, motion, etc.

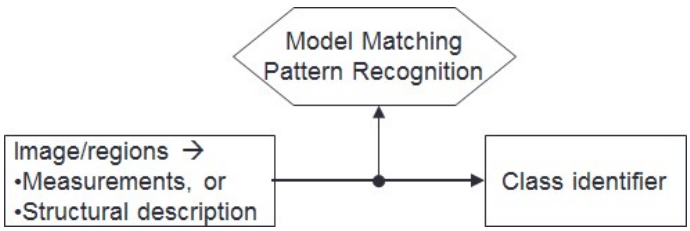
Image Analysis



Take useful measurements from pixels, regions, spatial relationships, motion, etc.:

- Greyscale/color intensity values.
- Size, distance.
- Velocity.

Pattern Recognition



Classify an image (region) into one of a number of known classes:

- Statistical pattern recognition (the measurements form vectors which are classified into classes).
- Structural pattern recognition (decompose the image into primitive structures).

Digital Image Representation

Image: 2D array of gray level or color values.

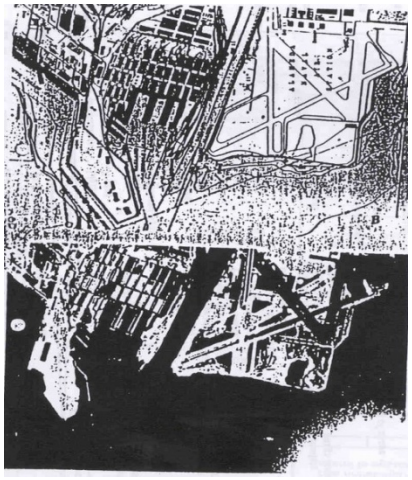
- Pixel: array element.
- Pixel value: arithmetic value of gray level or color intensity.

Gray level image: f=f(x,y)f = f(x,y)f=f(x,y)
3D image: f=f(x,y,z)f = f(x,y,z)f=f(x,y,z)
Color image (multi-spectral): f={Rred(x,y),Ggreen(x,y),Bblue(x,y)}f = \{R_{red}(x,y), G_{green}(x,y), B_{blue}(x,y)\}f={Rred(x,y),Ggreen(x,y),Bblue(x,y)}

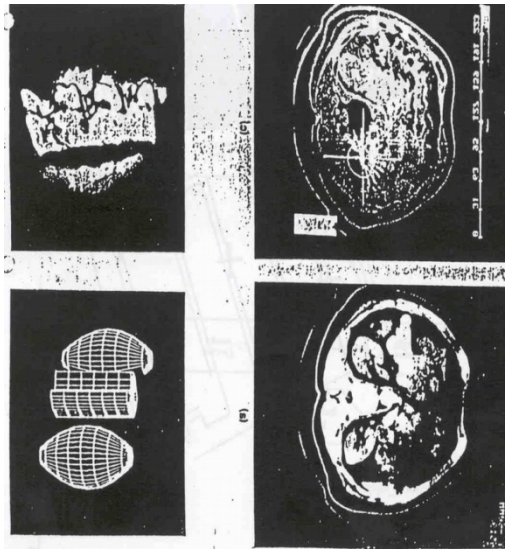
Machine Vision Applications

- Robotics
- Medicine
- Remote Sensing
- Meteorology
- Quality Inspection

Remote Sensing



- Take images from high altitudes (from aircraft, satellites).
- Find ships in the aerial image of the dock.
- Find if new ships have arrived.
- What kind of ships?

Medical Applications

- Assist a physician to reach a diagnosis.
- Construct 2D, 3D anatomy models of the human body.
- CG geometric models.
- Analyze the image to extract useful features.